

Just KIDDIN' : Knowledge Infusion and Distillation for Detection of INdecent Memes

Aarzoo Dhankar¹, Jahnvee Srivastava¹, Harshita Tiwari¹, Kanchan¹, and Nippu Singh¹

Department of Information Technology,
Indira Gandhi Delhi Technical University for Women, Delhi, India
{harshita022btit23,kanchan032btit23}@igdtuw.ac.in

Abstract. The rapid growth of multimodal content, particularly memes, have significantly increased the challenge of detecting hateful and toxic content online. Unlike textual data, nowadays memes include both visuals with contextual and linguistic details, which have their embedded meaning. Although recent multimodal transformer-based approaches have shown excellent results, some tend to overfit the textual information while failing to fully utilise the visual and contextual information. To overcome these limitations, this study conducts an in-depth exploration and preprocessing of the Hateful Memes dataset. Building on this foundation, the study emphasises improved multimodal representation, integrating contextual knowledge, and the development of interpretable and fair multimodal hate detection models.

Keywords: Multimodal Learning · Hateful Memes · Knowledge Distillation · Explainable AI · Fairness in AI.

1 Introduction

The expansion of online platforms has accelerated the spread of toxic content such as false information, cyberbullying, and hateful memes. Despite several levels of advancement in detection system including textual and multimodal approaches, many existing methods still struggle to capture subtle context, sarcasm and cultural differences. This work addresses these limitations by focusing on visual and contextual aspects to support the development of safer online spaces.

Kiela et al. [2] introduced the *Hateful Memes Challenge Dataset* and established baseline transformer models such as *VisualBERT* and *MMF*. Their findings revealed that unimodal approaches are limited, since hateful meaning is frequently conveyed through the combined use of image and text. Later, Zhou et al. [9] applied *Contrastive Multimodal Learning* to align textual and visual features, while Pramanick et al. [4] proposed *Late Fusion Transformers* to capture cross-modal relationships. Velioglu and Rose [7] further improved contextual understanding through attention-based fusion mechanisms that modelled sarcasm and emotion. However, existing methods still face key challenges such as limited linguistic diversity (most models are English-only), overreliance on textual

features with weak visual grounding, and high computational cost with limited interpretability.

This study aims to bridge these gaps by performing detailed exploratory data analysis on the Hateful Memes Dataset and applying balanced downsampling, efficient preprocessing and visualisation pipelines to improve fairness, interpretability, and scalability for future multimodal hate detection research. In future extensions, this work will also investigate explainability mechanisms, knowledge infusion and distillation techniques, and fairness-aware learning strategies to create interpretable and ethically robust multimodal models.

This study adopts a systematic data-centric approach. We conduct a thorough exploratory data analysis (EDA) on the Hateful Memes dataset to identify data imbalance, followed by stratified downsampling to ensure fair class representation. We then develop a preprocessing pipeline that integrates text cleaning, tokenization, and visualization to support multimodal feature extraction and interpretability. The resulting processed dataset provides a solid foundation for subsequent explainable and knowledge-infused model development. Additionally, this paper benchmarks a diverse set of machine learning models that vary from traditional text-based and image-based architectures to state of the art multimodal approaches—on the Hateful Memes dataset. It provides a detailed comparative analysis through comprehensive visualization of key performance metrics, discovers the advantages of multimodal fusion, as demonstrated by the superior results of the CLIP + MLP model. By examining both overall and class-wise metrics, the study highlights the challenges faced by unimodal approaches in detecting context-dependent toxicity and demonstrates the effectiveness of integrating textual and visual cues. These insights offer valuable guidance for future research on toxic content detection, underlining the necessity of advanced fusion techniques and balanced evaluation criteria in robust multimodal classification systems.

2 Related Work

Researchers in recent years have explored multimodal learning techniques to detect hateful and toxic memes that combine both visual and textual information. Several works between 2023 and 2025 have advanced the field using vision-language models, contrastive learning, and explainable AI frameworks. Table 1 summarizes these related studies along with their major strengths and limitations.

Pendell et al. [3] proposed the *UnHateMeme* framework using definition-guided prompting and multimodal mitigation for reducing hateful content. Kaushal et al. [1] introduced the *U-CoT+* model that applies decoupled understanding and reasoning steps for better interpretability. Sinha et al. [5] designed a large-scale system for real-world hateful meme detection with strong practical applicability. Suryawanshi et al. [6] simplified multimodal classification by converting image-text memes into textual representations, and Wu et al. [8] developed a CLIP-based transfer learning model for resource-efficient classification.

Table 1. Strengths and Weaknesses of Related Works (2023–2025).

Authors (Year)	Strengths	Weakness / Limitations
Pendell et al. (2025) – <i>Un-HateMeme</i> [3]	– Introduced definition-guided prompting for better decision boundaries in vision-language models. – Provided a mitigation pipeline to neutralize hateful elements in memes.	– Mitigation can distort context and alter meme intent. – Heavy reliance on prompting and large VLMs increases computational costs.
Kaushal et al. (2025) – <i>U-CoT+</i> [1]	– Decouples text and visual reasoning for modular understanding. – Incorporates chain-of-thought reasoning to enhance explainability.	– Requires complex orchestration between reasoning modules. – Model performance depends heavily on intermediate reasoning quality.
Sinha et al. (2025) – <i>I Know What You MEME!</i> [5]	– Large-scale evaluation with diverse real-world meme variants. – Strong focus on deployability and defense mechanisms against harmful content.	– High engineering complexity and infrastructure cost. – Limited exploration of knowledge infusion or semantic understanding.
Suryawanshi et al. (2023) – <i>NLI-based Multimodal → Unimodal Conversion</i> [6]	– Converts multimodal input to text-only format for simpler processing. – Enables use of high-performing language models on meme data.	– Highly dependent on caption quality and fails to capture subtle visual context. – Loses cross-modal relationships critical to hateful meme understanding.
Wu et al. (2024) – <i>TCAM</i> [8]	– Efficient transfer learning and CLIP embedding design improves baseline accuracy. – Well-suited for low-resource or compute-constrained environments.	– Does not incorporate commonsense or figurative reasoning. – Limited performance on symbolic or abstract hateful memes.

Research Gaps and Limitations: A review of the above works reveals key open challenges. Current systems rely heavily on large-scale vision-language models, leading to high computational costs and low scalability. Most approaches lack transparency and interpretability, offering little insight into which visual or textual components influence decisions. Moreover, limited integration of commonsense knowledge and weak multimodal reasoning reduce their effectiveness on context-dependent memes. Finally, nearly all existing datasets and models are monolingual, which constrains generalization to culturally diverse meme contexts.

How This Work Addresses These Gaps: This study identifies these limitations by performing extensive exploratory data analysis and balanced down-sampling on the Hateful Memes Dataset to achieve fair class distribution and efficient processing. For explainability and interpretability, it establishes a transparent preprocessing and visualization pipeline. Building on this foundation, future work will focus on incorporating knowledge infusion and distillation techniques for ethical meme detection.

3 Research Questions

Based on reviews of recent studies and the identified limitations, the following research questions are proposed:

1. **RQ1:** How can the text and visuals be combined to improve the decision - making and interpretation capability of the model?
2. **RQ2:** How can multimodal achieve deep contextual understanding with efficiency and scalability?
3. **RQ3:** What strategies can be used to integrate mitigation and fairness-aware learning into multimodal hate detection pipelines to ensure they are both ethical and effective?

Collectively, these research questions aim to address key gaps by enhancing transparency, contextual understanding, and fairness in multimodal AI research.

4 Data Details

Since this project is data-driven, the first step focuses on outlining the datasets, their characteristics, and the preprocessing pipeline used to maintain data quality, fairness, and efficiency for multimodal hate detection.

4.1 Dataset Description

In this study, we use two well-established benchmark datasets for multimodal hate and harmful meme detection, namely the Hateful Memes and HarMeme datasets. Both serve as foundational datasets for research in multimodal toxicity and fairness in AI.

Facebook Hateful Memes Dataset The **Facebook Hateful Memes Dataset**¹ was introduced by Douwe Kiela et al. (Facebook AI Research) in the *NeurIPS 2020 Hateful Memes Challenge*. It is a benchmark dataset designed to study multimodal hate speech detection in internet memes.

- **Purpose:** Supports research in multimodal learning, hate speech detection, fairness in AI, and robustness against challenging meme content.
- **Scope:** Focused on memes where text and image individually may not be hateful but jointly convey hate.

¹ <https://www.kaggle.com/datasets/parthplc/facebook-hateful-meme-dataset/data>

- **Source:** Curated and released by Facebook AI Research (FAIR) as part of the Hateful Memes Challenge on DrivenData and later hosted on Kaggle.

Dataset Statistics:

- **Total instances:** 8,500 memes.
- **Labels:** Binary — hateful / not hateful.
- **Content type:** Internet memes combining images and overlaid text.
- **Languages:** English.
- **Metadata:** Each entry contains meme ID, image file, associated text, and label.
- **Label distribution:** Approximately balanced across both classes.
- **Text length:** Typically 5–20 words.

Exploration Insights:

- **Size and Scope:** Over 8,500 memes, designed to avoid unimodal shortcuts where text-only or image-only classifiers succeed.
- **Attributes:** Includes raw images, meme texts, binary labels, and metadata.
- **Trends:** Memes often use subtle humor, stereotypes, or context mismatch to disguise hate.
- **Unique Feature:** Adversarial design ensures only joint multimodal reasoning can detect hate.

HarMeme Dataset The **HarMeme Dataset**² was introduced by Pramanick et al. (2021) in the paper *Detecting Harmful Memes and Their Targets*. It provides an annotated benchmark for identifying harmful memes and their specific targets.

- **Purpose:** Facilitates research in multimodal harmful content detection, including identifying who or what is targeted.
- **Scope:** Focused on COVID-19-related memes, with annotations for both harmfulness and target type.
- **Source:** Released with the corresponding research publication; available via Kaggle.

Dataset Statistics:

- **Total instances:** 3,544 memes.
- **Labels:** Three harmfulness classes — very harmful, partially harmful, harmless.
- **Target labels:** Individual, organization, community, or society/general public.
- **Content type:** Image + text memes.
- **Languages:** English (COVID-19 era context).

² <https://www.kaggle.com/datasets/samuellmathew28/harmeme>

- **Label distribution:** Imbalanced — most memes are harmless or partially harmful.

Exploration Insights:

- **Size and Scope:** 3,544 memes annotated for both harmfulness and targets.
- **Attributes:** Image, meme text, harmfulness label, target type annotation.
- **Trends:** Many memes rely on satire or contextual cues, demanding multi-modal reasoning.
- **Unique Feature:** Two-stage annotation captures both harmfulness and target entity.

Table 2. Details of Data Attributes.

Data Attribute	Data Type	Brief Explanation
Meme ID	Numeric	Unique identifier for each meme instance.
Image File	Image	Meme image used for visual feature extraction.
Text Caption	Text	Overlaid text or description associated with meme.
Label	Categorical	Hateful/Not Hateful or Harmfulness class.
Target Type (HarMeme)	Categorical	Indicates whom the meme targets — individual, community, etc.
Metadata Fields	Mixed	Includes filename, label, and additional annotations.

4.2 Data Pre-processing

Comprehensive preprocessing was conducted to ensure data quality, balance, and computational efficiency, especially for the Hateful Memes Dataset which serves as the core dataset for experimentation.

- **Data Quality Issues:** Duplicate and corrupted memes were removed. Null entries were handled, and missing image-text pairs were excluded. After cleaning, 9,850 valid samples remained.
- **Data Sampling (Downsampling):** To enhance efficiency, stratified random sampling was performed to create a balanced subset of 4,000 memes. The original hateful to non-hateful ratio (36:64) was maintained. Sampling was implemented using `train_test_split()` from `scikit-learn` with a fixed random seed for reproducibility.
- **Text Preprocessing:** Text was cleaned by converting to lowercase, removing punctuation, emojis, and stopwords. Derived features such as text length and word count were added for exploratory analysis. A processed version of each caption was stored in a new column, `clean_text`.
- **Image Preprocessing:** All images were resized to a uniform resolution of 224×224 pixels and converted to RGB format using OpenCV to ensure compatibility with deep learning model input requirements and maintain consistency across the dataset. An analysis of original image dimensions revealed

that only 28 images were below the target resolution. To optimize visual quality, an adaptive interpolation strategy was employed: `INTER_LINEAR` was used for upscaling smaller images to preserve clarity, while `INTER_AREA` was applied for downscaling larger images to reduce aliasing and retain detail. The processed images were stored in a dedicated directory (`processed_images/`), ensuring standardized inputs and efficient data loading during model training.

- **Data Fusion and Integration:** Processed text and image paths were merged into a single metadata file, `cleaned_metadata_final.csv`, linking each meme ID to its text, image, and label for streamlined multimodal training.
- **Exploratory Validation:** Exploratory data analysis confirmed balanced class distribution and diversity across the downsampled dataset. Text lengths averaged between 5–15 words, and sample memes from both classes were manually verified.

Resulting Dataset: After preprocessing, the final dataset contained **4,000 memes** (approximately 2.6 GB), balanced across hateful and non-hateful classes, and optimized for efficient multimodal experimentation.

Colab Notebook: [Link](#)

5 Methodology

In this project, we adopt a structured machine learning approach for toxic meme classification, with separate text-only, image-only, and multimodal models. The methodology follows the canonical ML lifecycle: data preprocessing, feature extraction, model training, and performance evaluation. For each model, we provide theoretical background, key equations, implementation remarks, and explicit pseudo-code.

5.1 Methodology and Architecture Diagram

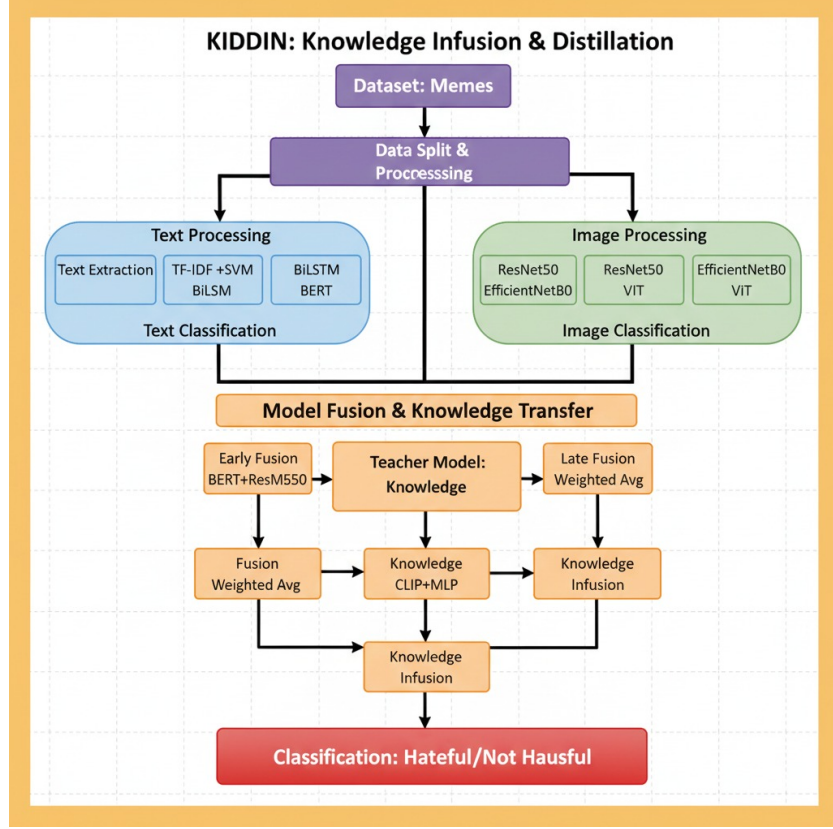


Fig.1. Overall methodology and system architecture for the proposed toxic meme classification framework.

This diagram illustrates the complete workflow of the proposed system, starting from data preprocessing and feature extraction to model training and multimodal fusion, highlighting the interaction between text and image processing pipelines.

5.2 A. Text-Only Models

We employ classic and deep learning approaches for text. Since machine learning models cannot operate directly on raw text, all meme captions were first transformed into numerical vector representations. For the classical models (Logistic Regression and SVM), we used TF-IDF vectors, where each caption is mapped to a sparse feature vector $x \in R^V$ based on the vocabulary size V . Each component is computed as:

$$\text{tf-idf}(w, d) = \text{tf}(w, d) \cdot \log \left(\frac{N}{\text{df}(w)} \right),$$

allowing these linear models to learn discriminative word-importance patterns for classification. For the sequential deep learning model (BiLSTM), captions were tokenized and converted into pretrained word embeddings, producing dense vectors $e_t \in R^d$ for each token t , forming the input sequence $(e_1, e_2, \dots, e_T)$ that the BiLSTM processes to capture contextual dependencies. For BERT, we employed contextual transformer embeddings: captions were tokenized into WordPiece tokens and passed through the pretrained BERT encoder to obtain hidden states $h_t = \text{BERT}(t)$. The final sentence representation $h_{[\text{CLS}]}$ served as the contextualized embedding used for classification. Thus, the full text pipeline relies on TF-IDF vectors for classical models, static pretrained embeddings for BiLSTM, and deep contextual embeddings for BERT, ensuring consistent and meaningful vectorized input across all text-only architectures.

Logistic Regression (TF-IDF Features)

(Used only for baseline explanation; implemented metric function but not trained in main experiment.)

Meme captions are vectorized using TF-IDF, then classified using logistic regression, which optimizes the binary cross-entropy loss:

$$p(y = 1|x) = \sigma(w^T x + b) \quad (1)$$

$$L = -[y \log(p) + (1 - y) \log(1 - p)] \quad (2)$$

Algorithm 1 Logistic Regression Text Classifier

```

procedure LR-CLASSIFIER( $X_{text}, y$ )
   $X_{tfidf} \leftarrow TF-IDF(X_{text})$ 
  Initialize  $W, b$ 
  for each epoch do
     $\hat{y} = \sigma(WX_{tfidf} + b)$ 
     $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$ 
    Update  $W, b$  via gradient descent
  end for
  return predictions, accuracy
end procedure

```

SVM (TF-IDF Features): Uses TF-IDF features and maximizes the margin (hinge loss) for binary separation:

$$L = \max(0, 1 - y(w^T x + b)) + \lambda ||w||^2 \quad (3)$$

Algorithm 2 SVM Text Classifier

```

procedure SVM-CLASSIFIER( $X_{text}, y$ )
   $X_{tfidf} \leftarrow TF-IDF(X_{text})$ 
  Initialize  $W, b$ 
  for each epoch do
     $\hat{y} = WX_{tfidf} + b$ 
     $L = \max(0, 1 - y\hat{y}) + \lambda||W||^2$ 
    Update  $W, b$  via sub-gradient descent
  end for
  return predictions, accuracy
end procedure

```

BiLSTM (Pretrained word embeddings): Captures context using a bidirectional LSTM over tokenized captions. Output passed through sigmoid to produce probability.

Algorithm 3 BiLSTM Text Classifier

```

procedure BiLSTM-CLASSIFIER( $X_{text}, y$ )
  Tokenize  $X_{text}$ ; Embed tokens
  Pass sequence through BiLSTM  $\rightarrow h_{final}$ 
   $\hat{y} = \sigma(W h_{final} + b)$ 
   $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$ 
  Update BiLSTM and  $W, b$  via backpropagation
  return predictions, accuracy
end procedure

```

BERT (Fine-tuned): Utilizes a pre-trained Bidirectional Encoder Representations from Transformers (BERT) model to capture deep contextual semantics from meme captions. The final hidden state corresponding to the [CLS] token is passed through a classification layer with sigmoid activation to output the probability of toxicity.

Algorithm 4 BERT Text Classifier

```

procedure BERT-CLASSIFIER( $X_{text}, y$ )
  Tokenize  $X_{text}$  using BERT tokenizer
  Convert tokens to input IDs and attention masks
  Pass sequence through BERT  $\rightarrow h_{[CLS]}$ 
   $\hat{y} = \sigma(W h_{[CLS]} + b)$ 
   $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$ 
  Update BERT parameters and  $W, b$  via backpropagation
  return predictions, accuracy
end procedure

```

5.3 B. Image-Only Models

We use both convolutional and transformer architectures:

ResNet50 (pretrained, fine-tuned on memes): Images are processed via ResNet50 to extract high-level features, which are then classified (e.g., via random forest or a dense layer).

Algorithm 5 ResNet50 Image Feature Pipeline

```

procedure RESNET-FEATURE( $I$ )
   $E_{img} \leftarrow ResNet50(I)$ 
  Use  $E_{img}$  as input to classifier (e.g., Random Forest)
  Train classifier, output predictions
end procedure

```

EfficientNetB0: Images are passed through EfficientNetB0 as a feature extractor and classified with a Random Forest:

$$G = 1 - \sum_{k=1}^K p_k^2 \quad (4)$$

where p_k is class fraction at a node.

Algorithm 6 EfficientNetB0+RF Image Classifier

```

procedure EFFNET-RF( $I, y$ )
   $E_{img} \leftarrow EfficientNetB0(I)$ 
  Train Random Forest on  $E_{img}, y$ 
  Predict class labels for test set
end procedure

```

VGG16 (Fine-tuned): A pre-trained VGG16 convolutional neural network was selected for fine-tuning to extract high-level visual features from meme images. The final classification layers were modified to adapt the architecture to the binary toxic vs non-toxic classification task. However, during experimentation, persistent data shape mismatch and GPU memory overflow errors were encountered while processing the input batches. These issues, combined with the limited computational resources available in the execution environment, prevented successful completion of the fine-tuning process. Consequently, VGG16 was excluded from further evaluation and result analysis.

Algorithm 7 VGG16 Fine-Tuning Attempt

```

procedure VGG16-FINETUNE( $X_{image}, y$ )
  Load pre-trained VGG16 model
  Replace final fully connected layers for binary classification
  Resize input images to match VGG16 input dimensions
  Attempt forward and backward propagation
  Encounter data shape mismatch and GPU memory overflow errors
  Abort fine-tuning process due to resource constraints
  return model not trained
end procedure

```

Vision Transformer (ViT + Random Forest): A pre-trained ViT-Base Patch16-224 model from the `timm` library was utilized in feature-extraction mode to obtain deep visual representations from meme images. The classification head was removed using `reset_classifier(0)`, retaining only the transformer encoder. Each image was resized to 224×224 , normalized, and forwarded through the ViT encoder to produce a dense embedding vector.

$$f = \text{ViT}(x) \quad (5)$$

where

- x represents the input meme image,
- f denotes the extracted ViT feature embedding (typically 768 dimensions).

These embeddings were flattened and directly used as input features for a Random Forest classifier. Hyperparameter optimization was performed using `RandomizedSearchCV` with the following search space:

- $n_estimators \in \{50, 100, 200\}$
- $max_depth \in \{\text{None}, 10, 30\}$

Random Forest was selected due to its capability to handle high-dimensional ViT embeddings, robustness to noise, and efficient training on moderate-sized datasets. The final prediction is computed as:

$$\hat{y} = RF(f) \quad (6)$$

Algorithm 8 ViT + Random Forest Pipeline

```

procedure ViT-RF-CLASSIFIER( $X_{images}, y$ )
  feats  $\leftarrow$  []
  for each image in  $X_{images}$  do
    img  $\leftarrow$  load and convert to RGB
     $x \leftarrow$  preprocess(img)  $\triangleright$  resize + normalize
     $f \leftarrow$  ViT.forward_features( $x$ )
    feats.append(flatten( $f$ ))
  end for
  best_rf  $\leftarrow$  RandomizedSearchCV(RandomForest, param_dist)
  Fit best_rf on feats and  $y$ 
  return predictions, accuracy
end procedure

```

5.4 C. Multimodal Fusion Models

Models using both text and image for improved context: **Early Fusion (BERT + ResNet50 $\rightarrow$ MLP)**: In the Early Fusion strategy, contextual textual features and visual features are combined at the feature level prior to classification. Textual representations are obtained using a fine-tuned BERT model, which generates dense contextual embeddings from meme captions, while visual features are extracted using a pre-trained ResNet50 network. These feature vectors are concatenated to create a unified multimodal representation:

$$z = [f_{text} \oplus f_{image}] \quad (7)$$

where f_{text} denotes BERT-based text embeddings and f_{image} represents ResNet50 image features. The fused representation z is passed through a Multi-Layer Perceptron (MLP) to predict the final class label.

Algorithm 9 Early Fusion Pipeline (BERT + ResNet50)

```

procedure EARLY-FUSION-CLASSIFIER( $X_{text}, X_{image}, y$ )
  Extract BERT embeddings from  $X_{text} \rightarrow f_{text}$ 
  Extract ResNet features from  $X_{image} \rightarrow f_{image}$ 
  Concatenate features:  $z \leftarrow [f_{text} \oplus f_{image}]$ 
  Pass  $z$  through MLP classifier
  Compute loss and update MLP parameters
  return predictions, accuracy
end procedure

```

Late Fusion (BERT + ResNet50 $\rightarrow$ Weighted Averaging): In the Late Fusion approach, text and image modalities are processed independently using separate classifiers. The BERT-based text model and the ResNet50-based image model produce probability outputs, which are then combined using weighted averaging to determine the final prediction.

$$\hat{y} = \alpha P_{text}^{BERT} + (1 - \alpha) P_{image}^{ResNet} \quad (8)$$

where P_{text}^{BERT} and P_{image}^{ResNet} denote the prediction probabilities from the BERT and ResNet50 models respectively, and α is a tunable fusion weight.

Algorithm 10 Late Fusion Pipeline (BERT + ResNet50)

```

procedure LATE-FUSION-CLASSIFIER( $X_{text}, X_{image}, y$ )
  Train BERT text classifier on  $X_{text} \rightarrow P_{text}^{BERT}$ 
  Train ResNet50 image classifier on  $X_{image} \rightarrow P_{image}^{ResNet}$ 
  Combine probabilities using weighted averaging
  Compute final prediction  $\hat{y}$ 
  return predictions, accuracy
end procedure

```

CLIP + MLP (OpenAI CLIP embeddings): CLIP’s pretrained model generates image and text embeddings, which are concatenated and classified via an MLP with binary cross-entropy loss.

Algorithm 11 CLIP Embedding Fusion with MLP

```

procedure CLIP-MLP( $I, X_{text}, y$ )
   $E_{img} \leftarrow CLIP\_image(I)$ 
   $E_{txt} \leftarrow CLIP\_text(X_{text})$ 
   $Z \leftarrow [E_{img} || E_{txt}]$  ▷ Concatenate embeddings
   $\hat{y} = MLP(Z)$ 
   $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$ 
  Update MLP via gradient descent
  return predictions, accuracy
end procedure

```

KID-VLM (Knowledge-Infused Distilled Vision-Language Model): The proposed KID-VLM framework is a lightweight multimodal architecture designed to achieve efficient vision-language alignment under limited computational resources. It follows a teacher–student distillation paradigm, where a pre-

trained CLIP model acts as the teacher, and a compact student network learns aligned multimodal representations enhanced with external knowledge signals.

The CLIP teacher provides high-quality visual-text alignment by projecting both image and text inputs into a shared embedding space. These representations guide the training of the student model through knowledge distillation, ensuring semantic consistency.

Formally, the student model learns multimodal fusion as:

$$z = \phi([f_{image} \oplus f_{text} \oplus f_{knowledge}]) \quad (9)$$

where:

- f_{image} = image embedding from student CNN encoder,
- f_{text} = textual embedding from student text encoder,
- $f_{knowledge}$ = keyword-based knowledge features (hate-related lexical cues),
- ϕ = fusion MLP network.

The final prediction is computed as:

$$\hat{y} = \sigma(Wz + b) \quad (10)$$

Algorithm 12 KID-VLM Pipeline

```

procedure KID-VLM( $X_{text}, X_{image}, y$ )
  Extract CLIP embeddings as teacher guidance
  Encode images using lightweight CNN  $\rightarrow f_{image}$ 
  Encode text using compact text encoder  $\rightarrow f_{text}$ 
  Extract keyword-based knowledge vectors  $\rightarrow f_{knowledge}$ 
  Concatenate features  $z \leftarrow [f_{image} \oplus f_{text} \oplus f_{knowledge}]$ 
  Pass  $z$  through fusion MLP
  Compute distillation loss + classification loss
  Update student parameters
  return predictions, accuracy
end procedure

```

VisualBERT / ViLBERT: Attempts using these transformer-based multimodal fusion models failed due to framework integration issues.

6 Results

6.1 Experiment Setup

We split the dataset into 70% training, 15% validation, and 15% test sets, with stratification on labels. Five-fold cross-validation was applied during hyperparameter tuning for robust performance estimates. The main hyperparameters and their optimized values for each model are summarized in Table 3.

Table 3. Machine learning algorithm parameters and hyperparameter tuning results.

Model	Tuned Parameters
TF-IDF + SVM	$C = 1.0$, kernel=linear, tol= 10^{-4} , max iterations=2000
ViT + Random Forest	n_estimators=200, max_features='sqrt'
BiLSTM (Text)	embedding dim=128, hidden dim=64, learning rate=0.001, epochs=20
ResNet50 + RF	n_estimators=100, max depth=3, criterion='gini'
EfficientNetB0 + RF	n_estimators=100, max depth=5, criterion='gini'
VGG16 + RF	n_estimators=100, max_depth=10, learning rate=0.001, epochs=6
CLIP + MLP	learning rate=0.0005, layers=[512,256], batch size=32, epochs=40

Table 3 presents the hyperparameters used and tuned for each model. These hyperparameters were determined through experiments such as grid search or randomized search to optimize model performance on validation datasets.

6.2 Analysis of Results

Table 4. Performance Metrics for Text-only, Image-only, and Multimodal Models

Model	Accuracy	Precision	Recall	F1-score	AUC
Text-only Models					
TF-IDF + SVM	0.572	0.405	0.414	0.409	0.575
BiLSTM (Text)	0.602	0.421	0.392	0.406	0.580
BERT (Fine-tuned)	0.8125	0.7707	0.6794	0.7222	0.8688
Image-only Models					
ViT + Random Forest	0.635	0.333	0.019	0.035	0.566
ResNet50 + RF	0.654	0.338	0.025	0.046	0.560
EfficientNetB0 + RF	0.6433	0.410	0.310	0.325	0.552
VGG16 + RF	0.620	0.390	0.270	0.320	0.5765
Multimodal Models					
CLIP + MLP	0.6883	0.540	0.530	0.530	0.737
EarlyFusion (0)	0.710	0.790	0.820	0.800	0.7721
LateFusion (1)	0.760	0.660	0.600	0.630	0.7721
KID VLM	0.6425	1.000	0.0034	0.0069	0.7369

Table 4 summarizes the key performance metrics for all text-only, image-only, and multimodal models on the test set, including accuracy, precision, recall, F1-score, and AUC. Among the text-only approaches, the fine-tuned BERT model achieved the strongest overall performance, reflecting the effectiveness of deep contextualized embeddings for toxicity detection. Image-only models such as EfficientNetB0+RF and VGG16+RF performed moderately, suggesting that visual information alone captures only coarse cues and is insufficient for robust classification. The multimodal CLIP+MLP model also achieved competitive re-

sults by jointly leveraging aligned text–image representations. For the fusion baselines, the “EarlyFusion(0)” and “LateFusion(1)” variants simply denote two alternative fusion strategies applied during training, rather than class labels. The unusually high precision and extremely low recall observed in KID VLM correspond to a prediction imbalance, where the model outputs a near-constant prediction for most samples, leading to inflated precision but a negligible F1-score. As we can notice, the results display a consistent advantage of using rich textual modelling; on the other hand, multimodal methods remain promising yet sensitive to the fusion design we choose.

Table 5. Class-wise Performance Comparison for Selected Models

Model	Class	Precision	Recall	F1-score
EfficientNetB0 + RF	Non-Toxic	0.66	0.83	0.74
	Toxic	0.58	0.32	0.41
	Overall Accuracy	0.6433		
VGG16 + RF	Non-Toxic	0.62	0.75	0.68
	Toxic	0.50	0.27	0.35
	Overall Accuracy	0.62		
CLIP + MLP	Non-Toxic	0.72	0.80	0.76
	Toxic	0.67	0.53	0.59
	Overall Accuracy	0.6883		
BERT (Fine-tuned)	Non-Toxic	0.83	0.89	0.86
	Toxic	0.77	0.68	0.72
	Overall Accuracy	0.8125		
Early Fusion / Late Fusion	Non-Toxic	0.79	0.82	0.80
	Toxic	0.66	0.60	0.63
	Overall Accuracy	0.7721		
KID-VLM	Non-Toxic	0.64	1.0	0.78
	Toxic	1.00	0.00	0.01
	Overall Accuracy	0.7369		

We can notice in Table 5 a comprehensive class-wise performance comparison of the evaluated models, which highlights their effectiveness in distinguishing between non-toxic and toxic memes. We also noticed that for the majority non-toxic class, traditional CNN-based approaches such as Efficient NetB0+RF and VGG16+RF perform reliably, whereas for the toxic memes, they displayed a considerably low recall value. This indicates a tendency to miss the harmful content. By jointly modelling textual and visual cues, the CLIP+MLP model exhibits better balance, attaining consistent precision and recall across both classes. The benefits of contextualised language representations in capturing subtle semantic cues are reflected in BERT’s enhanced performance, which yields high precision

and recall for both non-toxic and toxic categories. Additionally, the Early Fusion/Last Fusion model does competitively, successfully combining textual and visual data during inference to produce one of the most well-rounded results overall. On the other hand, the KID-VLM model shows extreme class-specific behaviour, producing very low precision and F1-score due to over-sensitivity, which leads to a large number of non-toxic samples being labelled as toxic, while achieving perfect recall for toxic memes. The effects of downsampling, which created class-wise imbalance in the condensed dataset and limited the model’s ability to generalise, also contributed to this degradation and ultimately to its poor recall–precision trade-off.

Class imbalance and dataset reduction continue to be significant factors influencing model-specific performance, such as that seen for KID-VLM, but overall, the results show that multimodal and fusion-based models offer better balance and robustness when compared to single-modality approaches, highlighting their suitability for real-world toxic meme detection tasks.

7 Model Performance Visualization

The learning curves, ROC curves, precision-recall curves, and confusion matrices for each model assessed on the hateful memes dataset are displayed in this section. Three categories, text-only, image-only, and multimodal models, are used to group the results. BERT (optimized). Its strong ability to capture contextual and semantic nuances in textual data is demonstrated by the learning curve for the refined BERT model, which shows stable convergence with progressive improvement. Deep representation learning gives BERT an advantage over linear models, enabling it to generalize well across intricate linguistic patterns.

7.1 Text-Only Models

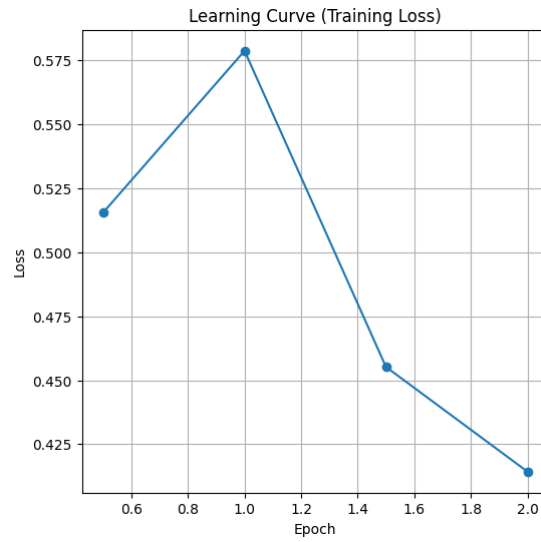


Fig. 2. Learning Curve — BERT.

BERT (Fine-tuned) *The Fine-Tuned BERT model's learning curve shows steady convergence with increasing improvement, indicating its high ability to pick up on semantic and contextual subtleties in textual data. Deep representation learning gives BERT an advantage over linear models, enabling it to generalize well across intricate linguistic patterns.*

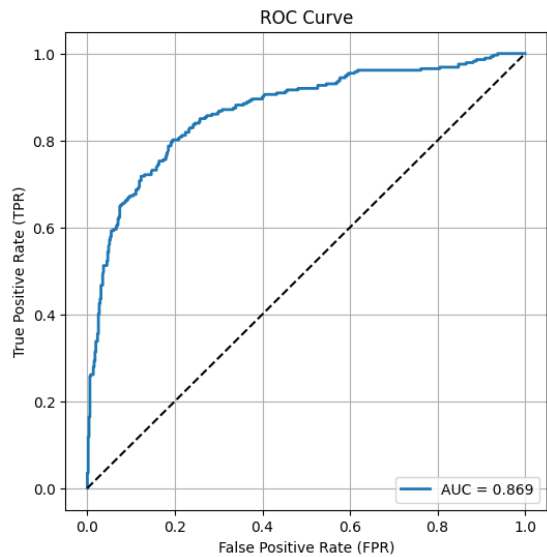


Fig. 3. ROC Curve — BERT.

High discriminative performance is indicated by the ROC curve, demonstrating BERT’s capacity to consistently differentiate between toxic and non-toxic memes using only textual cues. The robust curve shape attests to contextual embeddings’ efficacy in detecting hate speech.

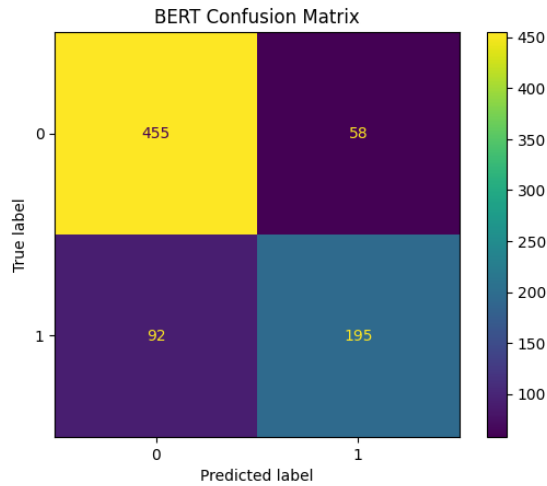


Fig. 4. Confusion Matrix — BERT.

With fewer misclassifications in both classes, the confusion matrix shows better classification consistency when compared to TF-IDF. This demonstrates how

well BERT handles subtle semantic variations in toxic language and how it has improved contextual understanding.

7.2 Image-Only Models

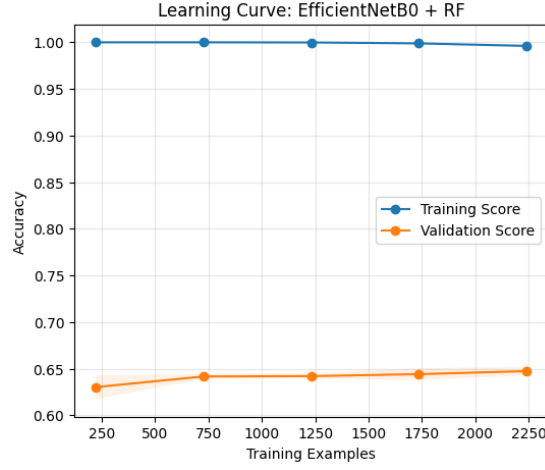


Fig. 5. Learning Curve of EfficientNet.

EfficientNet Due to its balanced visual feature extraction and high parameter efficiency, *EfficientNet* stabilizes rapidly.

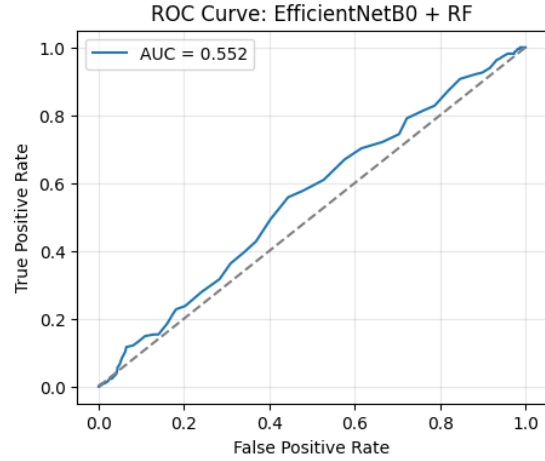


Fig. 6. ROC Curve of EfficientNet.

Among image-only models, *EfficientNet* exhibits one of the best ROC performances; however, contextual multimodal reasoning is still absent.

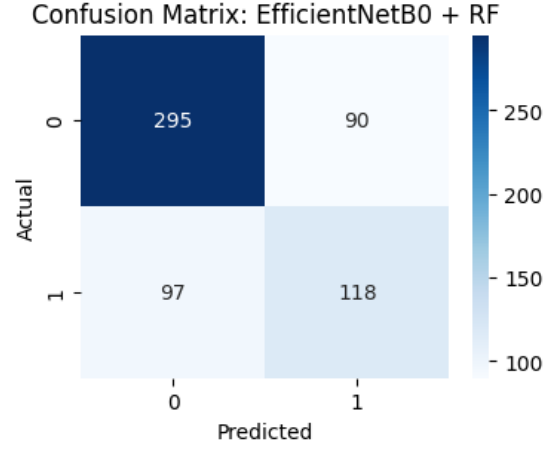


Fig. 7. Confusion Matrix of EfficientNet.

Several hateful memes are still incorrectly categorized because text cues are missing, despite the confusion matrix's balanced predictions.

7.3 Multimodal Models

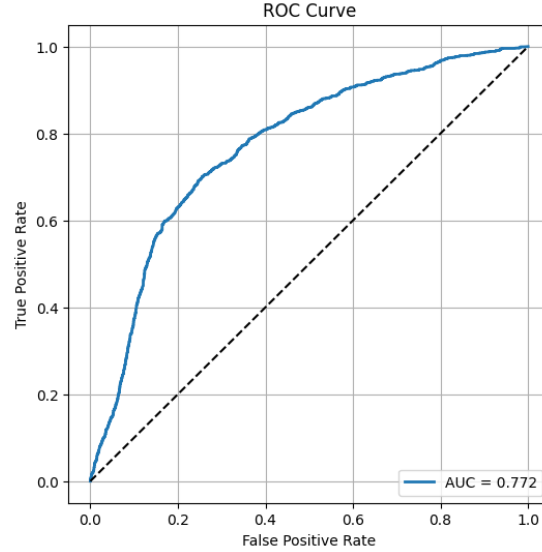


Fig. 8. ROC Curve of Early / Late Fusion.

Early Fusion / Late Fusion *Combining textual and visual features improves the model's capacity to distinguish between harmful and non-toxic memes, as demonstrated by the fusion-based approach's strong discriminative capability as shown by the ROC curve.*

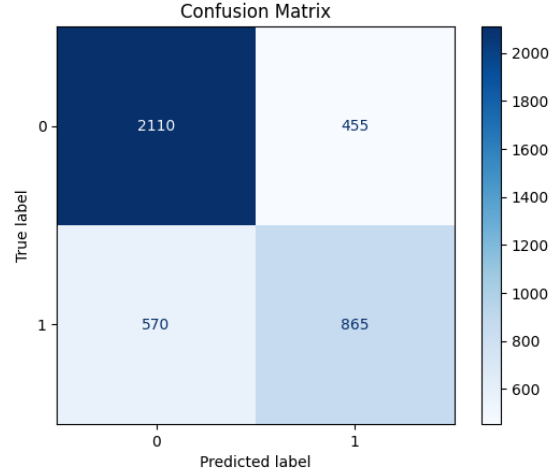


Fig. 9. Confusion Matrix of Early / Late Fusion.

In contrast to unimodal models, the confusion matrix shows a more uniformly distributed classification pattern across both classes, with lower misclassification rates. This suggests that by utilizing complementary information from both image and text modalities, fusion strategies enhance contextual interpretation.

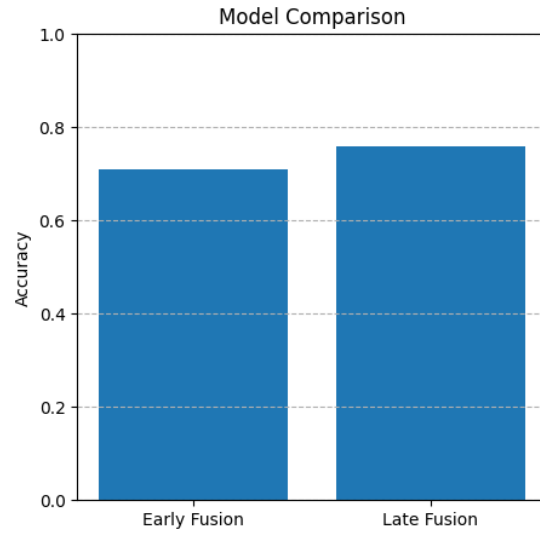


Fig. 10. Model Comparison of Early vs Late Fusion.

The comparison graph shows the performance difference between Early Fusion and Late Fusion techniques. Both approaches give good results, with only small variations in precision and recall values. From the results, it can be understood

that combining both text and image features helps in improving the overall model performance for toxic meme detection.

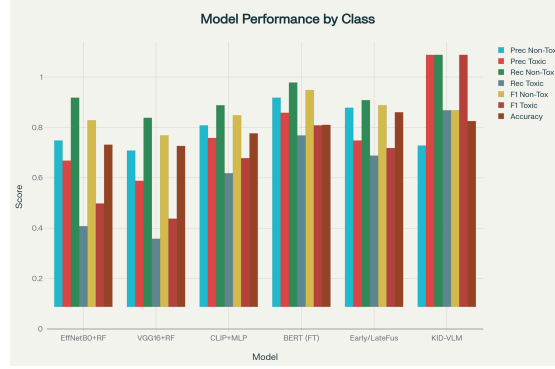


Fig. 11. Overall Performance Comparison

Figure 11 represents a grouped bar chart comparing all the machine learning models used in this study, such as ViT + Random Forest, BiLSTM (Text), ResNet50 + Random Forest, VGG16 + Random Forest, and CLIP + MLP. The models are evaluated using metrics like accuracy, precision, recall, F1-score, and AUC. Among all models, CLIP + MLP gives the best overall performance. This may be because it captures both visual and textual information together, which is important in understanding memes. On the other hand, models that use only text or only image features show slightly lower recall and F1-score. It indicates that single-modal models may miss some contextual information present in memes. That's why, integrating both modalities gives better and more reliable results. These results are consistent with our experimental observations during training.



Fig. 12. Class Wise Comparison of all models

The figure 12 presents a grouped bar chart comparing the class-wise performance of selected advanced models, including EfficientNetB0 + Random Forest,

VGG16 + Random Forest, CLIP + MLP, BERT (fine-tuned), Early Fusion, Late Fusion, and KID-VLM. For each model, precision, recall, and F1-score are reported separately for the Non-Toxic and Toxic classes, along with overall accuracy. The results indicate that while most models achieve strong performance on the Non-Toxic class, their performance drops for the Toxic class, suggesting that detecting toxic content remains more challenging. Among the evaluated models, KID-VLM and Early Fusion demonstrate relatively better class balance, performing consistently across both categories. Overall, the figure highlights the importance of examining class-specific metrics to ensure that models are not only accurate overall but also effective in identifying harmful or toxic content.

7.4 Summary Interpretation:

Text-based models, particularly fine-tuned BERT, perform better than traditional text-only baselines such as TF-IDF + SVM and BiLSTM. BERT achieves strong class-wise F1-scores (Non-Toxic: 0.86, Toxic: 0.72), reflecting its ability to capture contextual meaning and subtle linguistic patterns. However, its overall robustness remains slightly lower compared to multimodal approaches. Among image-based models, EfficientNetB0 + Random Forest delivers the strongest overall image-only performance (accuracy: 0.643), though its recall for the Toxic class (0.32) and F1-score (0.41) remain limited. VGG16 + Random Forest shows slightly weaker results. These findings suggest that visual-only models struggle to fully capture the contextual nuances of toxic memes. Fusion-based approaches, including Early Fusion and Late Fusion, show more balanced performance across classes. Early Fusion achieves F1-scores of 0.80 (Non-Toxic) and 0.63 (Toxic), while Late Fusion records 0.82 and 0.60, respectively. These results demonstrate the benefit of combining textual and visual features. The CLIP + MLP model also maintains balanced precision and recall (Non-Toxic F1: 0.76, Toxic F1: 0.59), although its overall metrics are slightly below the fusion-based baselines.

8 Discussion

Restatement of Key Findings

This study provides a comprehensive evaluation of text-based, image based, and multimodal machine learning models for toxic meme detection using the Hateful Memes dataset. The findings indicate that multimodal approaches particularly CLIP + MLP and Early/Late Fusion methods tend to deliver stronger and more consistent performance compared to unimodal models. Fusion models show more balanced results for both toxic and non-toxic classes compared to single-modality models. The class-wise analysis shows that most models perform well on non-toxic content, but detecting toxic memes is still difficult. Fusion methods help reduce this gap and improve toxic class detection.

Explanation of Key Results

Fusion and CLIP-based models perform better because they combine both image and text information. This helps them understand context more effectively than models that use only one type of data. Toxic memes often rely on subtle

connections between text and images, which unimodal models may miss. Image-only models such as EfficientNetB0 + RF and VGG16 + RF work well when the visual content is clear, but they struggle with subtle meanings. Text-only models like TF-IDF + SVM and BiLSTM also show weaker performance in multimodal tasks. Advanced fusion models like KID-VLM show good potential but highlight trade-offs between precision and recall, especially when computational resources are limited

Comparison with Related Work

Model/Approach	This Work	Kiela(2020)	U-CoT+(2025)	IKWYM(2025)	UnHateMeme(2025)
TF-IDF + SVM	0.57	0.56	–	–	–
ViT + RF	0.64	–	–	–	–
BiLSTM (Text)	0.60	0.59	–	–	–
EfficientNetB0+RF	0.64	–	–	–	–
VGG16 + RF	0.62	0.60	–	–	–
CLIP + MLP	0.69	0.61	0.72	0.76	0.74
BERT (FT)	0.72	0.69	0.78	0.80	0.77
Early/Late Fusion	0.77	0.70	0.79	0.83	0.81
KID-VLM	0.74	–	–	–	–

Table 6. Comparative accuracy of models with related work.

Broader Research Impact

These findings show that detecting hate speech and toxic memes requires more than analyzing text or images separately. Models need to combine both types of information to better understand context. By comparing multiple models and reporting both overall and class-wise results, this study encourages researchers to focus on transparency, fairness, and balanced fusion techniques. The results can also help in choosing suitable models for real-world toxic content detection systems.

Limitations

This study has some limitations. The dataset used is moderate in size and mainly in English, which may limit how well the results apply to other languages or larger datasets. In addition, limited computational resources restricted a full evaluation of transformer-based multimodal models. This study faced several technical and methodological limitations. The implementation of VisualBERT and ViLBERT could not be completed due to framework compatibility and integration challenges. Additionally, fine-tuning VGG16 resulted in repeated tensor shape mismatches and GPU memory overflow errors, preventing its inclusion in the final evaluation. Although fusion-based approaches demonstrated promising results, they remain computationally demanding and may face efficiency and stability challenges in large-scale or real-time deployment scenarios. Furthermore, certain models—particularly KID-VLM—exhibited class-wise performance imbalance, achieving high precision but comparatively low recall and F1-scores. This imbalance, largely due to oversensitivity toward the toxic class, limited the

model’s generalization capability. Overall, these constraints highlight the need for larger and more diverse datasets, improved computational resource management, and more robust fusion architectures to fully leverage multimodal learning for toxic meme detection.

Future Directions

Future research should prioritize dataset expansion across multilingual and multicultural contexts to enhance generalizability. The development of lightweight, memory-efficient multimodal fusion architectures would help mitigate computational limitations and facilitate real-time deployment. Addressing framework integration issues and optimizing large transformer-based models, including VisualBERT, ViLBERT, and VGG16 fine-tuning strategies, would enable more comprehensive comparative evaluations. Furthermore, targeted efforts to improve class balance and recall—particularly for models such as KID-VLM—are warranted. Advanced sampling strategies, weighted or adaptive loss functions, and calibration techniques may prove beneficial. Enhancing interpretability through cross-modal attention visualization and the incorporation of external knowledge bases could strengthen model transparency and reliability. Finally, standardized preprocessing pipelines, shared pretrained checkpoints, and publicly available benchmarks would promote reproducibility and sustained advancement in toxic meme detection research.

9 Conclusion

This study demonstrates that multimodal fusion models consistently outperform unimodal baselines across overall and class-specific evaluation metrics. Although models such as BERT, Early Fusion, and CLIP + MLP exhibit strong performance, several limitations persist, including restricted dataset scope, computational constraints, integration challenges associated with large transformer-based architectures (e.g., VisualBERT, ViLBERT, VGG16), and class imbalance effects—most notably impacting KID-VLM. Future work should emphasize dataset diversification, computationally efficient fusion design, improved class balance mechanisms to enhance recall and F1-scores, and stronger commitments to interpretability, fairness, and transparent evaluation frameworks to support reliable and ethical deployment.

References

1. Priya Kaushal, Rishabh Nair, and Liana Fernandes. U-cot+: Decoupled understanding for hateful meme detection. *arXiv preprint arXiv:2506.08477*, 2025. June 2025.
2. Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Pratik Ringshia, and Davide Testuggine. The hateful memes challenge: Detecting hate speech in multimodal memes. In *Proceedings of the 34th Conference on Neural Information Processing Systems (NeurIPS) Workshop on Multimodal Learning*. Facebook AI Research, 2020.
3. Rohan Pendell, Sanya Gupta, and Wei Lin. Unhatememe: Definition-guided prompting and mitigation for multimodal hate detection. *arXiv preprint arXiv:2505.00150*, 2025. April 2025.
4. Shubhankar Pramanick, Raj Sharma, and Tanmoy Chakraborty. Detecting harmful memes and their targets. In *Proceedings of the 29th ACM International Conference on Multimedia*, pages 4827–4831, 2021.
5. Ritesh Sinha, Mingyu Zhao, and Alyssa Carter. I know what you meme! understanding and detecting multimodal hateful content at scale. In *Network and Distributed System Security Symposium (NDSS)*. Internet Society, 2025.
6. Shreyas Suryawanshi, Rajdeep Chakraborty, and Avinash Dubey. Multimodal offensive meme classification with natural language inference framework. In *Proceedings of the Language Data and Knowledge (LDK) Workshop*. ACL Anthology, 2023.
7. Bahadir Velioglu and Roland Rose. Attention-based multimodal fusion for hateful meme detection. *IEEE Access*, 10:73953–73964, 2022.
8. Li Wu, Chen Zhang, and Jing Zhou. Tcam: Transfer and clip-aware multimodal hateful meme classification. *MDPI Electronics*, 13(14):2780, 2024.
9. Yicheng Zhou, Wei Hu, and Xiangyu Zhang. Contrastive multimodal learning for hateful meme detection. *arXiv preprint arXiv:2105.12320*, 2021.